

Online Appendix

to

Workforce Strain in the Production of Accounting

Estimates:

Evidence from CECL Adoption

Haozheng Wang

Zicklin School of Business, Baruch College, CUNY

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OA1. Guide to the Online Appendix

This Online Appendix documents the construction of the data used in the paper and reports checks and supplementary analyses that support the interpretation of the main results. Sections OA2 and OA3 describe the regulatory panel, CECL adoption-date review, employer matching, review collection, and aggregation. Section OA4 describes the recruitment-posting evidence and the occupational classification of employee job titles. Section OA5 presents the review-text taxonomy and its implementation. Section OA6 reconciles the review-level records with the BHC-quarter measures used in the regressions. Section OA7 contains supplementary empirical analyses. Variable definitions appear in Appendix A of the paper.

The descriptions below distinguish the inputs used to construct the measures from information used only to develop or assess the classification rules. Historical employee reviews are classified from their self-reported job titles and review text. Recruitment postings help interpret occupational boundaries, but no historical reviewer is assigned duties from a later job posting. The regression outcomes and CECL adoption status are not supplied to either language-model classification task.

OA2. Regulatory Data and Panel Construction

OA2.1. FR Y-9C panel and target institutions

The regulatory panel begins with quarterly FR Y-9C filings for U.S. bank holding companies (BHCs). The focal estimation period is 2015Q1 through 2024Q4. Filings through 2025Q4 provide the subsequent net charge-offs required for observations near the end of the focal period. The download retains the balance-sheet, income-statement, credit-loss, loan-composition, and employment fields used in the paper. The panel is keyed by BHC RSSD identifier and calendar quarter.

Glassdoor collection targets BHCs whose total consolidated assets reach at least \$5 billion

in the regulatory panel. For each qualifying BHC, I use the WRDS Bank Structure database to identify U.S. commercial-bank subsidiaries and the corresponding ultimate parent. The resulting search file contains one row for the parent name and one row for each distinct commercial-bank subsidiary. The current file contains 663 parent–subsidiary name rows representing 334 BHCs.

OA2.2. CECL adoption-date review

I initially identify CECL adoption from the first quarter in which BHCKJJ26, BHCKJJ27, or BHCKJJ28 is nonmissing. A reported zero is treated as valid reporting evidence. I then review the preliminary date against annual reports, SEC filings, and other public disclosures. The review file records the BHC, the preliminary date, the action taken, the final date when applicable, the entity to which the evidence pertains, the document and page, the relevant wording, the source link, and explanatory notes.

The current review file contains 133 BHC-level records. For 111 records, the final treatment date is set from the documented review; 15 retain the regulatory-panel date; and seven are treated as never adopting within the observed BHC entity history. The last category includes cases in which a later adoption disclosure pertains to a successor entity and cannot be assigned to the former BHC in the panel. The treatment indicator equals one beginning in the reviewed adoption quarter and zero beforehand.

OA2.3. Heterogeneous-loan share

The quarterly heterogeneous-loan share equals construction and land-development loans, commercial real estate loans, and commercial and industrial loans divided by total loans and leases held for investment. The construction first uses the legacy Schedule HC-C Part I M fields. Construction and land development equals BHCKM708 + BHCKM710

+ BHCKM712; commercial real estate equals BHCKM714 + BHCKM716 + BHCKM719; commercial and industrial loans equal BHCKM727 + BHCKM729 + BHCKM731; and total loans equal BHCKM746 + BHCKM748 + BHCKM750. When the required M fields are unavailable, the construction uses the corresponding Schedule HC-C Part II items BHCKJJ04, BHCKJJ05, BHCKJJ07, and BHCKJJ11. Components must be jointly observed and non-negative, the denominator must be positive, and the resulting share must lie between zero and one. Missing components are not replaced with zero.

For the cross-sectional analysis, I average the valid quarterly shares over the eight quarters preceding the reviewed adoption date, requiring at least three available quarters. This pre-adoption measure is fixed over the BHC’s estimation period. *HighShare* equals one when the measure is at or above the BHC-level median in the relevant comparison-group sample. *PreSize* is the average of unadjusted log assets over the same eight-quarter window, also requiring at least three available quarters and remaining fixed during estimation.

OA3. Glassdoor Employer Matching and Review Collection

OA3.1. Employer-name matching

Glassdoor reviews may appear on a page associated with either the parent BHC or a subsidiary bank. I therefore search the platform using every parent and subsidiary name in the regulatory control file. Each legal name produces a light variant that removes legal and geographic suffixes and an aggressive variant that also removes structural banking terms. The procedure preserves short brand names, retains periods in abbreviations, rejects suffix stripping that would leave a stem shorter than four characters, and creates an additional “Bank” variant for eligible single-word stems.

The first search round evaluates the light, original, and aggressive variants in that order and examines up to three candidates per query. Candidate names are compared using

character, token-sort, and token-set similarity. An accepted match must satisfy a minimum score of 95, a query-to-candidate length ratio of at least 0.50, and compatibility between the first substantive tokens. First tokens may match exactly or through a common prefix of at least four characters. Queries of five characters or fewer require a character-based score of at least 95. Additional restrictions reject cases in which a high token-set score is produced by a much longer candidate name.

The second round applies additional punctuation, parenthetical, regional, and appended-“Bank” transformations to names not matched in the first round. It examines up to ten candidates. The general threshold is 92, increasing to 94 for aggressive variants and 95 for variants that remove a regional qualifier. The minimum length ratio is 0.40. Short-name, first-token, and superset protections remain in place. Generic variants such as “First Bank” or “Community Bank” are not searched when they result from aggressive stripping. Two parent–page links are added after manual verification. Names that do not satisfy these procedures remain unmatched.

When multiple subsidiary names under the same BHC resolve to the same Glassdoor page, I retain one parent–page link. The same page may remain linked to more than one BHC when the mapping file supports both links. A physical review on such a page enters once for each linked parent but never more than once within a parent. This treatment preserves the observed legal-entity mapping rather than assigning a shared page to an arbitrary parent.

OA3.2. Review collection, extraction, and integrity checks

For each matched page, I collect the available review history. The raw files contain review identifiers, posting dates, job titles, employee status, location, structured ratings, and the written *Pros*, *Cons*, and advice fields. Existing review files from a prior project are reused only when their earlier integrity records indicate a successful download. Remaining pages

Table OA1: Glassdoor Matching and Review-Pipeline Reconciliation

Pipeline stage	Records	BHCs	Glassdoor pages
Regulatory parent–subsidiary search file	663	334	—
First-round accepted matches	401	255	304
Second-round accepted matches	51	37	35
Final parent–page mapping after manual additions and within-parent deduplication	365	284	338
Pages with usable review files	—	281	331
Physical reviews in the collected history	430,218	—	331
Physical reviews dated 2015Q1–2026Q2	386,401	—	330
Mapped review–parent observations dated 2015Q1–2026Q2	424,391	281	330

This table reconciles the current pipeline files. First- and second-round counts are search-file rows and can contain repeated pages or BHCs. The final mapping is unique by BHC and Glassdoor page. Seven of the 338 matched pages did not yield a usable nonempty JSON review file. Physical-review counts precede parent mapping. Mapped review–parent counts are higher because 23 Glassdoor pages are linked to more than one BHC. The number of pages in the 2015Q1–2026Q2 row is lower because one usable page contains no review in that period. The regressions separately restrict their index quarters and rolling review windows as described in Section OA6.

are collected directly, and each saved file must parse as a nonempty JSON list. Files that are absent, empty, or unreadable are placed on a rescrape list rather than silently entering the review panel.

The extraction step preserves each physical record in the raw file and assigns a stable source-row identifier. Text fields are converted to valid Unicode, internal whitespace is collapsed, and zero-valued structured ratings are treated as missing. Work/Life Balance, Overall, Career Opportunities, Compensation and Benefits, Culture and Values, and Senior Leadership remain on their one-to-five scales. Posting dates are converted to calendar quarters. The baseline construction retains reviews across reported employment statuses and locations. Employee status, U.S.-location verification, and job-title availability remain in the primitive data so that restricted-sample analyses can be constructed without recollecting reviews.

The extraction currently contains 430,218 physical review records from 331 Glassdoor pages. All records have usable posting dates. The extraction output identifies 902 additional records whose provider review identifier repeats within the same page. The baseline retains these physical records; sensitivity analyses can instead retain one record per page

and provider identifier.

OA4. Recruitment Postings and Occupational Classification

OA4.1. Recruitment-posting evidence

Employee reviews report job titles but generally do not describe the duties attached to those titles. I therefore collect U.S. job listings from the matched Glassdoor employer pages to inform the interpretation of title expressions. The inventory contains 26,529 unique job identifiers from 198 employer pages. The title and description scraper obtains usable descriptions for 26,527 postings. The corpus contains 16,731 distinct listing titles and 20,683 distinct nonempty descriptions.

The descriptions provide a reference corpus for interpreting occupational boundaries and documenting selected title expressions. They do not create a historical employee-level match, and the final review classifier does not impute a reviewer’s duties from a posting with the same or a similar title. No automated classification of posting descriptions enters the review classifications or empirical measures; specific posting examples require separate manual verification.

OA4.2. Fixed classification of the risk function

The review-title classifier operates in two stages. The first stage applies fixed deterministic rules to identify the risk group. Titles are lowercased, punctuation is replaced with spaces, and repeated whitespace is collapsed. For selected families, rank and edge modifiers such as “senior,” “associate,” “lead,” and numeric job grades are removed before testing the core occupational expression. Meaningful qualifiers remain available to the rules.

Table OA2 summarizes the ten rule families. The rules identify occupational functions, not confirmed participation in CECL estimation. A title can therefore enter the risk group

because it names a broad bank risk, quantitative, portfolio, research, economics, actuarial, stress-testing, or loss-forecasting function even when the title alone does not reveal the employee’s specific assignment.

OA4.3. Classification of the remaining functions

The second stage sends every remaining nonempty title to GPT-5.6 Luna.¹ The model receives only the job title and the fixed category definitions. It does not receive the employer, review date, ratings, review text, CECL status, or the output of other records. Each title is classified independently under a strict output schema. A missing title is assigned to *unknown* without a model call. A title that identifies an occupation outside the nine named functions is assigned to *other*; a title that does not resolve an occupation is assigned to *unknown*. The risk group cannot be selected in this stage.

The primary comparison group consists of titles assigned to the client-facing category. The broader non-risk group contains every review not assigned to the risk group, including reviews assigned to *other* or *unknown*. Missing and uninformative titles therefore enter the non-risk benchmark but never the client-facing benchmark.

The production program checks that every physical review receives exactly one function, that all missing-title records remain unknown, and that the risk indicator agrees with the risk category.

¹The production identifiers are `v4_rri_quant_reviewer_engineer` for the fixed risk-title rules and `nonrisk_functions_v1.3_fixed_risk_gate_single_no_id_v1` for the remaining-function prompt.

Table OA2: Fixed Risk-Function Title Rules

Rule family	Included expressions	Representative titles encoded in the rule file
Credit and loan assessment	Credit or loan analyst, analysis, manager, specialist, associate, modeling, administration, review, research, or committee; credit officer; selected qualified loan-officer expressions; credit risk, policy, or approval	Credit Analyst; Credit Associate; Consumer Loan Officer
Underwriting	Titles containing an underwriting expression	Underwriter; Mortgage Underwriter; Commercial Underwriting Manager
Model development and validation	Model risk, development, developer, modeler/modeling, and qualified model-validation expressions	Model Validator; Model Validation; Model Risk Analyst
Quantitative work	Quantitative titles combined with analysis, analytics, research, engineering, development, strategy, modeling, management, or related occupational nouns	Quantitative Analyst; Quantitative Researcher; Quantitative Analytics Specialist
Portfolio work	Portfolio analyst, manager, associate, management, monitoring, or credit-portfolio expressions	Portfolio Manager; Portfolio Analyst; Credit Portfolio Consultant
Research	Exact research-analyst, equity-research, fixed-income-research, and credit-research expressions	Research Analyst; Equity Research Analyst; Equity Research Associate
Economics and actuarial work	Economist and actuarial expressions	Economist; Actuarial Analyst
Stress testing and regulatory frameworks	Stress-testing expressions and CCAR, DFAST, Basel, or ICAAP	Stress Testing Analyst; CCAR Analyst; Basel Specialist
Loss forecasting	Loss-forecasting, credit-loss-forecasting, and forecasting-analyst expressions	Credit Loss Forecasting Analyst; Forecasting Analyst
Other explicit risk roles	Occupational nouns located near “risk,” including broad enterprise, market, operational, business, technology, financial, and liquidity-risk roles	Risk Analyst; Operational Risk Manager; Market Risk Analyst; Risk Reviewer

The table summarizes the production rule file rather than reproducing every regular expression. A candidate match is excluded when the title instead identifies accounting, bookkeeping, audit, financial or regulatory reporting, tax accounting, or general-ledger and reconciliation work. Exact generic, mortgage, and commercial loan-officer titles, their rank variants, and assistant-to-loan-officer titles are excluded from the risk gate; a title such as Consumer Loan Officer is not covered by that exception. The first matching family provides the primary audit label, while all family matches are retained in the classification metadata.

Table OA3: Definitions for Non-Risk Occupational Functions

Function	Definition
Client-facing	Customer service, sales, advice, account and relationship management, origination, and management of customer-facing branches or markets. Examples include tellers, personal and universal bankers, relationship managers, financial advisors, loan officers, mortgage bankers, commercial lenders, and branch managers.
Markets	Trading, market execution, and advisory or execution work on corporate financial transactions. Trade settlement and processing are classified as operations.
Accounting and finance	Accounting, tax, financial and regulatory reporting, internal planning and budgeting, performance analysis, and management of the bank's own cash, funding, and liquidity.
Compliance	Regulatory compliance, AML, KYC, sanctions, financial-crime investigation, fraud detection and prevention, and related monitoring.
Audit	Independent audit and assurance and the design, evaluation, or testing of internal financial-reporting controls.
Operations	Transaction processing, settlement, clearing, servicing, collections, document handling, and routine operational administration.
IT and data	Software development, infrastructure, technical support, systems engineering, data analysis and science, data engineering, software testing, and technical cybersecurity.
Projects and transformation	Product, project, and program management; business, process, and requirements analysis; process improvement; change delivery; and transformation.
Corporate support	Human resources, recruiting, compensation and benefits, employee learning, legal counsel, marketing and communications, procurement, administration, facilities, physical security, and workplace services.
Other	An identifiable occupation outside the nine named functions, including clearly identified general leadership that lacks a more specific functional assignment.
Unknown	Missing, generic, conflicting, or otherwise insufficient occupational information. Rank labels such as Analyst, Associate, Vice President, Director, Manager, or Officer alone remain unknown.

The production prompt contains additional boundary cases and familiar-title defaults. It instructs the model to use *unknown* when two functions are equally plausible and to avoid inferring duties from a bank business area or seniority alone.

Table OA4: Distribution of Occupational Classifications

Function	Full collected history		2015Q1–2026Q2	
	Reviews	Percent	Reviews	Percent
Risk	14,325	3.33	13,146	3.40
Client-facing	104,369	24.26	92,546	23.95
Markets	4,649	1.08	4,288	1.11
Accounting and finance	11,888	2.76	10,682	2.76
Compliance	8,664	2.01	8,349	2.16
Audit	1,961	0.46	1,838	0.48
Operations	21,098	4.90	18,973	4.91
IT and data	40,541	9.42	38,075	9.85
Projects and transformation	20,031	4.66	18,337	4.75
Corporate support	13,027	3.03	12,011	3.11
Other	9,091	2.11	8,340	2.16
Unknown	180,574	41.97	159,816	41.36
Total	430,218	100.00	386,401	100.00

Counts are physical review records before mapping a shared Glassdoor page to multiple BHCs. Risk is determined by the fixed keyword stage. The remaining nonempty titles are assigned by GPT-5.6 Luna, and missing titles are assigned to *unknown*. The full history spans all posting dates in the collected files; the second pair of columns begins in 2015Q1 and ends at the last collected review in 2026Q2.

OA5. Classification of Written Employee Reviews

OA5.1. *Input, blinding, and output structure*

I classify the *Cons* field because it allows employees to choose the negative experiences they consider worth reporting. GPT-5.6 Luna receives the *Cons* text and a temporary record identifier.² It receives no employer, job title or function, date, rating, employee status, location, or CECL information. The prompt treats embedded instructions as review text and directs the model to classify only experiences stated or clearly implied by the review.

The taxonomy contains eight non-mutually-exclusive substantive categories. Each category is evaluated independently, so one review may receive several labels. A strict output schema requires a binary value for every category and every record. The model also identifies text with no substantive complaint. When no substantive category applies, the pipeline assigns the record mechanically to exactly one of two residual categories, *No Complaint* or *Other Complaint*. The classifier therefore does not select a single primary topic, and the category percentages need not sum to 100 percent.

OA5.2. *Operational distinctions*

Several boundaries are central to the classification. Excessive work amount, time demands, understaffing that increases work, or explicitly negative work-life balance enter *Workload*. Performance targets, scrutiny, accountability, and changing or conflicting expectations enter *Role and Accountability* only when the source of the demand is identifiable. A review can receive both labels when it separately describes both experiences. Generic pressure or stress does not establish either label.

Micromanagement and restricted discretion enter *Control and Autonomy*. Hostility, dis-

²The production prompt identifier is `taxonomy_v1.2_clarifications_1_2`.

Table OA5: Complaint Taxonomy and Coding Boundaries

Category	Code one when the review describes	Do not infer the category from	Illustrative excerpt
Workload and time demands	Excessive work amount or intensity, excessive time demands, long hours, unreasonable deadlines, insufficient staffing or capacity that raises work demands, workload-linked burnout, or explicitly negative work-life balance	Generic stress, pressure, targets, scrutiny, difficult judgment, inefficient processes, or micromanagement without work-amount, time, capacity, or work-life-balance content	“Sometimes too much work to be done in short time.”
Control and autonomy	Restricted discretion over how, when, or where work is performed, including micromanagement, rigid scheduling, or responsibility without authority	Rude or incompetent management, slow approvals, unclear expectations, bureaucracy, or heavy workload without a personal-control restriction	“Your time is micromanaged from your breaks to your lunch.”
Production frictions	Identifiable process, system, data, documentation, coordination, approval, handoff, or rework obstacles that make work harder to complete	Vague inefficiency, generic bureaucracy, interpersonal conflict, or poor leadership communication without an operational obstacle	“Lots of red tape and hard to get things done.”
Role and accountability demands	Unclear, conflicting, changing, high-stakes, unusually scrutinized, or explicitly demanding expectations, targets, responsibilities, or accountability	Work volume by itself, lack of autonomy with clear expectations, process problems alone, or generic stress without an identifiable source	“Difficult goals that change constantly and almost impossible to reach.”
Managerial and interpersonal support	Lack of managerial or coworker support, hostility, disrespect, abusive behavior, poor leadership relationships, or dysfunctional interpersonal communication	Micromanagement alone, workflow failures, or unfairness without an independent support problem	“Unorganized management, no guidance or leadership.”
Instability and insecurity	Layoffs, job-loss risk, disruptive restructuring, role elimination, merger disruption, or organizational instability that creates employment uncertainty	Turnover or organizational change without instability, disruption, or employment uncertainty	“Not knowing if you will have your job the next week.”
Rewards and career development	Pay, benefits, promotion, recognition, career paths, professional development, or training directed at long-run development	Employment insecurity or training problems that only prevent completion of current tasks	“Promotion is limited and can take years.”
Organizational justice	Unfair, inconsistent, discriminatory, preferential, retaliatory, biased, or arbitrary treatment or procedures	Politics, ethics concerns, low pay, lack of promotion, rude management, or opacity without an explicit fairness concern	“There is somewhat of a culture of favoritism.”
<i>No Complaint</i>	No meaningful negative complaint, including “none,” “no cons,” clearly positive text, or meaningless filler	Any substantive criticism	—
<i>Other Complaint</i>	A meaningful negative complaint is present, but none of the eight substantive categories applies	This category is constructed mechanically and cannot coexist with a substantive label or <i>No Complaint</i>	—

The excerpts are short examples from the review data. The production prompt contains additional inclusions, exclusions, and cross-category boundaries. A label requires direct textual evidence or a clear semantic implication; keywords and negative tone alone are insufficient.

respect, and lack of interpersonal support enter *Managerial and Interpersonal Support*. An identifiable process, system, data, approval, or coordination obstacle enters *Production Frictions*. Poor management by itself does not establish a production friction, and bureaucracy by itself does not establish a loss of employee control.

Training problems enter *Production Frictions* when they prevent completion of current work and *Rewards and Career Development* when they limit longer-run development. Turnover enters *Workload* only when the review describes the resulting work or staffing burden and enters *Instability and Insecurity* only when the review describes disruption or employment uncertainty. Turnover alone receives neither inference. Similar restrictions prevent a vague reference to politics, change, inefficiency, or hierarchy from being semantically completed into a substantive label.

OA5.3. *Execution and reconciliation*

The production classifier evaluates batches of 50 records with reasoning effort set to none. The response schema fixes one output slot per input record and rejects missing, additional, duplicated, or nonbinary outputs. Failed batches are not published as partial classifications. Completed responses and their configurations are checkpointed, and a changed prompt or changed input produces a different cache identity. The final production file contains 430,218 classified records.

The aggregation step verifies that all substantive and residual indicators share the same denominator, that positive counts never exceed the number of classified reviews, and that the residual categories obey the taxonomy. It then sums valid-label counts and positive labels within posting quarter, BHC, occupational function, employment status, geography group, and title-availability group. Rating sums and valid-rating counts are stored separately, so BHC-quarter means are calculated from individual ratings rather than from averages of subgroup means.

Table OA6: Distribution of Classified Complaint Categories

Complaint category	All reviews (%)	Risk function (%)	Client-facing (%)	Non-risk (%)
Workload and time demands	19.80	15.83	23.37	19.94
Control and autonomy	10.70	9.61	12.90	10.74
Production frictions	15.60	16.35	13.31	15.57
Role and accountability demands	12.86	9.61	19.64	12.98
Managerial and interpersonal support	20.71	18.21	24.90	20.80
Instability and insecurity	7.82	8.89	6.53	7.78
Rewards and career development	35.69	39.62	35.81	35.56
Organizational justice	9.26	8.05	9.24	9.30
<i>No Complaint</i>	8.89	7.94	6.96	8.93
<i>Other Complaint</i>	12.00	12.14	11.03	11.99
Classified reviews	430,218	14,325	104,369	415,893

The table uses physical reviews over the full collected history before parent mapping. The eight substantive categories can overlap. *No Complaint* and *Other Complaint* are mutually exclusive residual categories and receive no substantive label. The non-risk column contains every occupational category other than risk, including *other* and *unknown*.

OA5.4. *Separate understaffing classification*

The pipeline also contains a separate binary classification for explicit or clearly implied shortages of human resources. The label includes too few staff, unfilled positions or missing replacements, insufficient backup or shift coverage, denied headcount needed to perform the work, and too few qualified employees. It does not infer understaffing from heavy workload, long hours, layoffs, turnover, a small team, or limited resources unless the text identifies a staffing or coverage shortage. This classification uses only the *Cons* text and does not alter the eight-category taxonomy. In the full review history, 3.47 percent of reviews receive the understaffing label, compared with 1.91 percent of risk-function reviews and 6.29 percent of client-facing reviews.

OA6. Construction of BHC-Quarter Workforce Measures

OA6.1. *Posting-quarter primitives*

The review pipeline first creates posting-quarter primitives rather than final rolling measures. For each BHC, quarter, function, employee-status group, geography group, and title-

availability group, it records the number of physical reviews; the number and sum of valid ratings for each structured dimension; and the number of valid and positive text labels for each complaint dimension. This design allows the analysis code to reconstruct alternative populations without averaging precomputed means.

The baseline measures collapse across current and former employees, reported and unverified locations, and title availability. Risk and client-facing observations retain only reviews assigned to those functions. The broader non-risk group sums all functions other than risk, including *unknown*. When a Glassdoor page maps to several BHCs, its reviews contribute once to each mapped parent and the primitive file records the resulting review–parent observations.

OA6.2. Forward review window for H1

For a BHC-quarter indexed by t , the H1 window contains reviews posted in quarters t through $t + 3$. Rating means equal the sum of individual ratings divided by the number of nonmissing ratings within the group and window. The primary measure is

$$WLB_gap_{i,t} = \overline{WLB}_{i,t}^{\text{Risk}} - \overline{WLB}_{i,t}^{\text{Client}}. \quad (\text{OA1})$$

The alternative non-risk gap replaces the client-facing mean with the mean for all non-risk reviews. The WLB sample requires at least three total reviews and an observed WLB mean for every function used in the specification. Thus, the three-review condition concerns the number of reviews in the group, while the mean uses every nonmissing WLB rating among those reviews.

For text outcome c , the BHC-window complaint rate is

$$ComplaintRate_{i,t}^{c,g} = \frac{\sum_{j \in (i,t,g)} \mathbf{1}\{j \text{ is classified in } c\}}{N_{i,t,g}^{\text{classified Cons}}}. \quad (\text{OA2})$$

The primary complaint gap is the risk-function rate minus the client-facing rate; the alternative subtracts the broader non-risk rate. The text sample requires at least three classified *Cons* reviews for each group used. Rates and their differences are proportions, so 0.01 equals one percentage point.

For adopting BHCs, quarters -3 through -1 relative to adoption are excluded from H1 because their forward windows contain reviews posted in the adoption quarter or later. This restriction prevents a nominally pre-adoption outcome from using post-adoption reviews.

OA6.3. Historical review windows for H2

The allowance-informativeness analysis constructs prior risk-function and non-risk WLB from reviews posted in quarters $t - 4$ through $t - 1$. Each group must have at least three total reviews and an observed WLB mean. The prior window and the quarter- t allowance must lie entirely within the same accounting regime. The requirement removes the adoption quarter and the following three quarters because their historical review windows contain pre-adoption observations. Accounting regime is assigned from the quarter- t allowance; the baseline does not additionally require all quarters used to realize future net charge-offs to remain in that regime.

The conditional-change specification also constructs earlier risk-function WLB from quarters $t - 8$ through $t - 5$. Define

$$\Delta RiskWLB_{i,t} = Prior\ Risk\ WLB_{i,t} - Earlier\ Risk\ WLB_{i,t}. \quad (OA3)$$

A positive value indicates improvement in risk-function WLB between the earlier and recent windows. Both windows require at least three total risk-function reviews and an observed WLB mean. The earlier window supplies the conditioning baseline and may cross the adoption date. The regression includes the earlier level and its allowance interaction alongside

the change measure and its allowance interaction. This parameterization contains the same information as a model with the two WLB levels and their allowance interactions, while making recent improvement the focal coefficient.

OA7. Supplementary Analyses

OA7.1. Continuous heterogeneous-loan share

The main cross-sectional analysis uses an indicator for BHCs with an above-median pre-adoption share of heterogeneous loans. Table OA7 instead interacts *Post* with the underlying continuous share. This specification preserves the information in the distribution of loan composition and tests whether the main inference depends on the median split. The first two columns use client-facing employees as the comparison group, and the final two use the broader non-risk workforce. Columns (2) and (4) additionally allow the post-adoption change to vary with pre-adoption BHC size.

The coefficient on $Post \times Share$ is negative in all four specifications. With the client-facing comparison, it equals -0.528 without the size interaction ($p = 0.122$) and -0.612 when the size interaction is included ($p = 0.052$). With the broader non-risk comparison, the corresponding estimates are -0.611 ($p = 0.026$) and -0.649 ($p = 0.015$). Because the share is expressed from zero to one, a ten-percentage-point larger pre-adoption heterogeneous-loan share corresponds to a 0.053–0.065 point more negative post-adoption change in relative risk-function WLB across the four specifications. The direction is consistent with the indicator-based analysis, although the estimate without the size interaction in the client-facing sample is not statistically distinguishable from zero at conventional levels.

Table OA7: **Continuous Heterogeneous-Loan Share and Employee Working Conditions**

	<i>WLB_gap</i> (Client)		<i>WLB_gap</i> (Non-risk)	
	(1)	(2)	(3)	(4)
<i>Post</i>	−0.024 (0.279)	1.575* (0.918)	0.063 (0.222)	0.606 (0.824)
<i>Post</i> × <i>Share</i>	−0.528 (0.339)	−0.612* (0.312)	−0.611** (0.271)	−0.649** (0.261)
<i>Post</i> × <i>PreSize</i>	—	−0.088* (0.047)	—	−0.029 (0.044)
Observations	1,751	1,751	1,802	1,802
BHCs	111	111	117	117
Within R^2	0.051	0.058	0.041	0.042
Financial controls	Yes	Yes	Yes	Yes
BHC fixed effects	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes

This table replaces the above-median indicator in the main cross-sectional analysis with the continuous pre-adoption heterogeneous-loan share. The share is the average proportion of held-for-investment loans consisting of construction and land development, commercial real estate, and commercial and industrial loans over the eight quarters preceding adoption; it is expressed from zero to one. *PreSize* is the corresponding average of unadjusted log assets and is not centered. Columns (1) and (2) use *WLB_gap* based on client-facing employees; columns (3) and (4) use the broader non-risk benchmark. Columns within each comparison group use the same sample. All specifications include log assets, the capital ratio, the EBLP ratio, and the deposit ratio, together with BHC and calendar-quarter fixed effects. Because the moderators are not centered, the standalone coefficients on *Post* are evaluated where the included moderators equal zero and are not economically interpreted. Standard errors clustered by BHC appear in parentheses.

*, **, and *** denote two-sided significance at the 10, 5, and 1 percent levels, respectively.

OA7.2. *Loan composition and pre-adoption BHC size*

The joint specifications account for pre-adoption size because loan composition and BHC scale are related in the estimation samples. Table OA8 reports this relation using one observation per BHC. The two columns correspond to the samples supporting the client-facing and broader non-risk comparisons in Table OA7.

The Pearson correlations are −0.298 in the client-facing sample and −0.321 in the non-risk sample, and the rank correlations are similar. In the bivariate regressions, a one-unit increase in pre-adoption log assets is associated with a 3.6- to 3.8-percentage-point smaller heterogeneous-loan share. The estimates remain negative after including adoption-cohort fixed effects. These results document that the focal portfolio measure is related to pre-adoption scale in the estimation samples. They motivate allowing the post-adoption change

Table OA8: **Pre-Adoption Heterogeneous-Loan Share and BHC Size**

	Client-facing sample	Non-risk sample
Pearson correlation	−0.298***	−0.321***
<i>p</i> -value	0.0015	0.0004
Spearman correlation	−0.292***	−0.317***
<i>p</i> -value	0.0019	0.0005
OLS coefficient on <i>PreSize</i>	−0.0357***	−0.0384***
Robust standard error	(0.0102)	(0.0100)
R^2	0.089	0.103
OLS coefficient on <i>PreSize</i> , adoption-cohort fixed effects	−0.0420***	−0.0430***
Robust standard error	(0.0123)	(0.0118)
R^2	0.197	0.207
BHCs	111	117

The table uses one observation per BHC in the corresponding Table OA7 estimation sample. The heterogeneous-loan share and *PreSize* are BHC-level averages over the eight quarters preceding CECL adoption, each requiring at least three observed quarters. The share is expressed from zero to one; *PreSize* is unadjusted log assets. The first OLS specification is bivariate. The second includes fixed effects for the BHC's adoption quarter. OLS standard errors are heteroskedasticity robust. *, **, and *** denote two-sided significance at the 10, 5, and 1 percent levels, respectively.

in WLB to vary with *PreSize*; they do not assign a causal interpretation to the cross-sectional size–portfolio relation.

References